From Friction to Flows: An NLP-Based Feedback Review System for Optimizing Product Performance at Any.do

COURSE: COMPUTER SCIENCE FOR COLLECTIVE INTELLIGENCE III

SCHOOL OF COLLECTIVE INTELLIGENCE

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1. Introduction

Productivity applications are expected to function as invisible cognitive infrastructure, quietly supporting users as they plan tasks, manage time, and organize routines. Any.do, a task and project management application with over 40 million users (about twice the population of New York) worldwid**e**, is designed with this promise in mind. However, large-scale user feedback suggests that the application does not consistently meet this expectation.

User reviews reveal that friction often emerges at moments when focus and continuity matter most. Premium pop-up prompts interrupt task creation, essential features such as sequential reminders or advanced scheduling are locked behind paywalls, and reliability issues, particularly around synchronization, widgets, and notifications, disrupt daily workflows. These interruptions accumulate, turning what should be a seamless productivity experience into a fragmented one.

This problem is not trivial. According to the AppDynamics App Attention Span Study, more than 90% of users abandon applications that exhibit poor performance, intrusive design, or restrictive monetization models. In productivity tools, where switching costs are low and alternatives are abundant, repeated friction directly translates into user churn.

This project addresses the cost of a broken workflow by applying Natural Language Processing (NLP) within a collective intelligence framework. Rather than relying on isolated complaints or star ratings alone, we treat user reviews as a distributed, continuously updated knowledge base. By systematically analyzing sentiment and extracting recurring themes from thousands of reviews, the project aims to generate actionable insights that help Any.do realign its product roadmap with actual user needs and evaluate whether fixes reduce friction over time.

2. Background and Conceptual Foundations

2.1 NLP and Collective Intelligence

NLP, as we understand it, is a subfield of artificial intelligence that enables machines to process, understand, and generate human language. Its relevance to collective intelligence lies in its ability to extract structured insights from large volumes of unstructured text, allowing patterns of shared experience to emerge from individual expressions.

Customer feedback analysis using NLP has been widely adopted in both research and industry. Early work by Pang and Lee (2008) demonstrated the feasibility of sentiment classification for opinion mining, while later surveys by Liu (2012) highlighted a key limitation: numerical ratings and surface-level sentiment indicators often fail to capture nuanced dissatisfaction. In industry, companies such as Amazon and Google have acknowledged the value of text-based sentiment analysis for understanding customer experience, while also recognizing challenges such as sarcasm, mixed sentiment, and domain-specific language.

This project builds these insights by explicitly comparing rating-based sentiment assumptions with model-based sentiment inference, addressing a gap that persists in many applied sentiment analysis workflows.

2.2 Sentiment Analysis and Semantic Analysis

Sentiment analysis is commonly implemented as a text classification task that assigns polarity labels to text. While effective at scale, this approach can oversimplify user experience when sentiment is inferred solely from numerical scores.

Semantic analysis extends this capability by focusing on meaning, context, and word relationships. Techniques such as word frequency analysis, n-grams, TF-IDF weighting, and topic modeling allow analysts to move from "how users feel to why they feel that way". This layered approach reflects the course's emphasis on combining quantitative classification with qualitative interpretation to support informed decision-making.

3. Materials and Methods

3.1 Data Source and Scope

The Any.do Google Play reviews (Kaggle dataset) consist of **16,092 Google Play Store reviews** for the Any.do application, each containing columns: review ID, free-text feedback (content), a numerical rating (score), and a timestamp (at). Figure 1 (Score Distribution) shows that 3-star reviews dominate the dataset, highlighting a large neutral segment that traditional analyses often overlook.

To support clear polarity detection, reviews with a neutral score (3) were removed for the binary sentiment workflow. Scores of 4–5 were mapped to positive sentiment, and scores of 1–2 to negative sentiment. After filtering and deduplication, **11,034 unique reviews** formed the core dataset for modeling and analysis.

3.2 Computational Environment

All analyses were conducted in the Google Collab using Python. Data processing relied on pandas and NumPy, text normalization used regular expressions and lemmatization tools, semantic modeling used scikit-learn and gensim, and sentiment classification was performed using **pysentimiento**, a pretrained transformer-based model from the Hugging Face ecosystem.

4. NLP Pipeline and Execution

4.1 Phase 1: Data Cleaning and Preparation

The initial dataset shape was **(16,092 × 4)**. Duplicate reviews were removed using **reviewId**, reducing redundancy and ensuring that repeated submissions did not artificially inflate certain opinions. Text was normalized by removing special characters, converting to lowercase, and lemmatizing words to their base forms. This ensured that semantic patterns reflected meaning rather than noise.

4.2 Phase 2: Exploratory Data Analysis (EDA)

EDA revealed several critical patterns supported by the attached figures:

- **Sentiment Distribution:** Using score-based classification, Any.do appears generally healthy, with **4,475** positive reviews for scores of 4–5 outweighing **3,423** negative ones for scores of 1–2, alongside **3,136** neutral reviews for those with score 3. This is shown below **(Figure 1 and 2).**

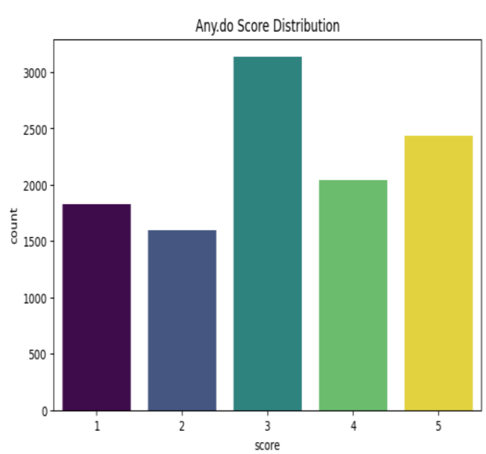

Figure 1: Any.do review score Distribution

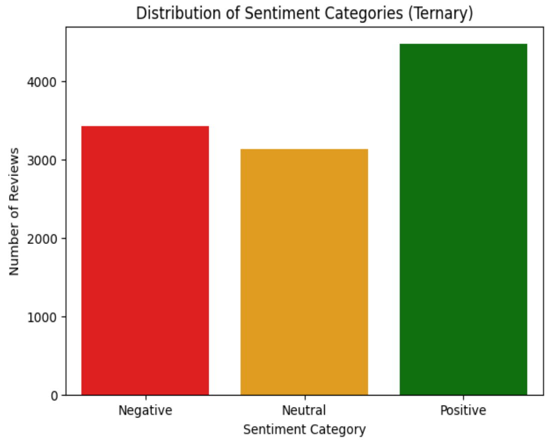

Figure 2: Sentiment Categories Distribution

- **The Neutral "Swing" Segment:** Neutral reviews represent a large, engaged population that is neither fully satisfied nor disengaged, an important opportunity for retention.
- **Temporal Trends (Figure 3):** From 2014 to 2018, review volume and sentiment were low and stable. A sharp surge in early 2020 saw positive reviews exceeding 1,400 per month, coinciding with a major growth phase. Importantly, negative reviews also rose sharply above 400, indicating scaling-related friction.

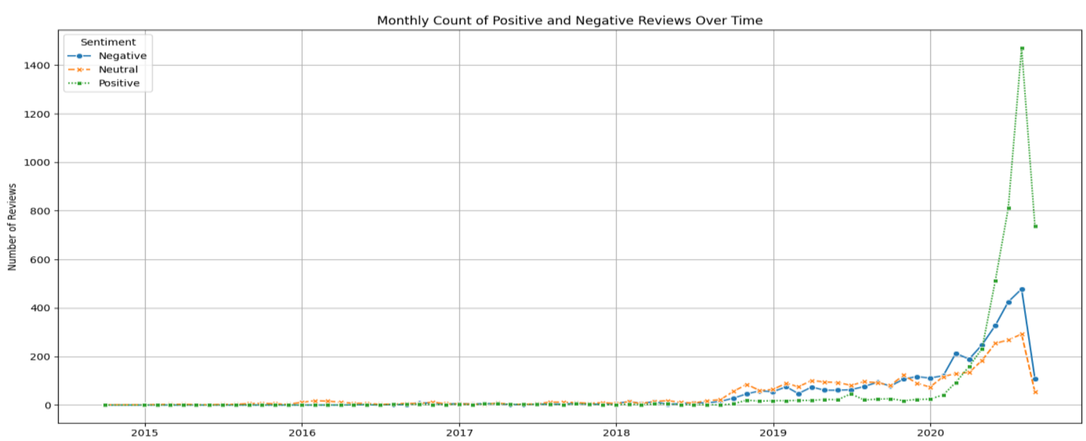

Figure 3: Distribution of sentiment (positive, neutral and negative reviews over time)

- **Word Clouds (Figure 4 & 5):** Positive reviews emphasize “tasks,” “calendar,” and “list,” confirming strong core functionality. Negative reviews concentrate on “premium,” “ads,” “sync,” “doesn’t,” and “can’t,” pointing to monetization and reliability failures.

These patterns established the need for deeper sentiment and semantic modeling.

Figure 4: World cloud (positive reviews)

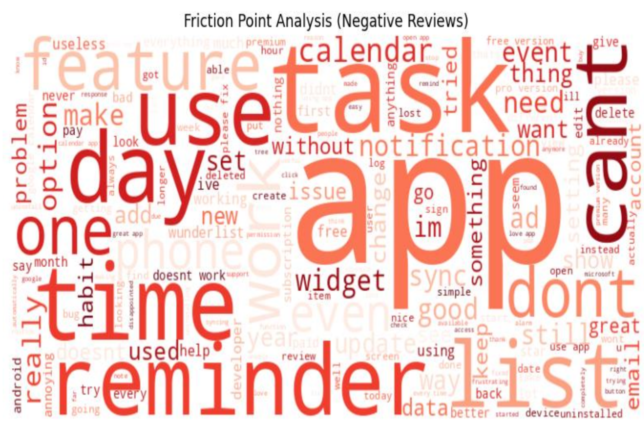

Figure 5: World cloud (negative reviews

4.3 Phase 3: Sentiment Classification with pysentimiento (Hugging Face Model)

We used pysentimiento, a pretrained Hugging Face model finetuned for sentiment analysis, to classify customer feedback in content column efficiently without custom training. Reviews were reclassified based on linguistic content rather than numerical scores. The resulting ternary sentiment distribution was:

- **Positive:** 4,512
- **Neutral:** 3,439
- **Negative:** 3,083
- This result is visualized in the final sentiment pie chart showing that over 58% of reviews fall into the Neutral or Negative categories when analyzed textually. Crucially, this step revealed high-score reviews that nonetheless expressed negative sentiment, exposing a major limitation of rating-based analysis. This is shown below in **(Figure 6)**

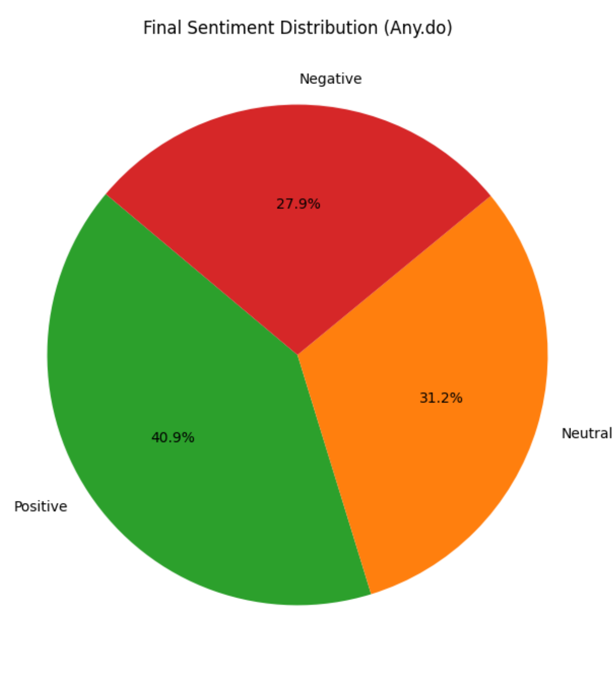

Figure 6: Overall Sentiment Distribution of Any.do User Feedback

4.4 Phase 4: Pain Point Detection and Topic Modeling

Negative reviews were grouped and analyzed using word frequency analysis, n-grams, TF-IDF weighting, and Latent Dirichlet Allocation (LDA). Topic modeling revealed three dominant frustration themes:

1. **Monetization Friction and Stability:** Complaints link premium or paid versions with crashes and poor performance, indicating resentment when financial investment is not matched by reliability.
2. **Core Productivity Workflow Failures:** Issues with reminders, tasks, lists, and calendars directly undermine the app's fundamental purpose.
3. **Feature Reliability and Update Regressions:** Widgets and notifications frequently stop working after updates, highlighting weaknesses in regression testing and background service stability.

These themes move beyond surface complaints to identify structural drivers of dissatisfaction.

4.5 Phase 5: Insights and Reporting

Aggregating these findings through a collective intelligence lens allows prioritization based on consensus rather than isolated feedback. The analysis identifies synchronization stability and premium feature reliability as the most urgent threats to user retention. The neutral segment represents a high-value group that could be converted into advocates if these friction points are addressed.

5. Results and Discussion

A central contribution of this project lies in demonstrating the limitations of quantitative, score-based sentiment analysis. While numerical ratings suggest a broadly positive user base, NLP-based sentiment modeling reveals a more nuanced reality: users often assign high ratings while expressing dissatisfaction in text.

The pysentimiento model, leveraging contextual language understanding, captures negation, frustration, and mixed sentiment that numeric scores alone obscure. This qualitative sensitivity reduces the risk of misinformation in product evaluation and highlights why NLP-enhanced collective intelligence is essential for informed decision-making.

By integrating sentiment classification with semantic analysis, the project shows not only "how users feel, but what specifically drives those feelings", enabling evidence-based prioritization rather than intuition-driven fixes.

6. Ethical Considerations and Bias Assessment

The analysis uses publicly available data and operates at an aggregate level, minimizing privacy risks. However, selection bias remains inherent, as reviewers tend to have stronger opinions than silent users. Model bias is also possible, as pretrained sentiment models inherit biases from their training corpora. These limitations are acknowledged explicitly, and findings are treated as indicative signals rather than absolute truths.

7. Conclusion and Future Directions

This project demonstrates how NLP and collective intelligence can transform raw user feedback into a strategic asset. By moving beyond star ratings and leveraging transformer-based sentiment models and semantic analysis, the study provides a clear, evidence-based roadmap for improving Any. Do's user experience.

Future work should extend this pipeline to aspect-based sentiment analysis, incorporate continuous monitoring through retrieval-augmented systems, and evaluate whether implemented changes measurably shift sentiment over time. As demonstrated, NLP is not merely a technical tool, but a decision-making framework that enables organizations to listen to their users at scale, accurately, responsibly, and continuously.

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